A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives

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Contents

Chapter 1 **Research Summary** — Page 2

1.1 Introduction 2

1.2 Meta-analysis of Related Literature 3

Overall Trend — 3 • Quantitative Analysis — 3 • Methodology Review — 4

Chapter **Acknowledgment** — Page 6

Chapter **References** — Page 7

Chapter 1

Research Summary

Preface

This report provides a comprehensive summary and critical analysis of *A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives* by Zheng et al. The following sections synthesize the authors' framework, methodology, and key findings regarding Agricultural Parcel and Boundary Delineation (APBD)

1.1 Introduction

Definition 1.1.1: Agricultural Parcel and Boundary Delineation

Agricultural Parcels represent the fundamental spatial units for agricultural activity.

Agricultural Parcel and Boundary Delineation (APBD) is an interdisciplinary application of computer algorithms for vision tasks where agricultural fields are disected into their component parcels, and distinctly separated from their background (adjacent parcels, natural vegetation and other types of land).

Applications of APBD include:

- sustainable agriculture planning and land-use monitoring
- crop mapping
- yield estimation

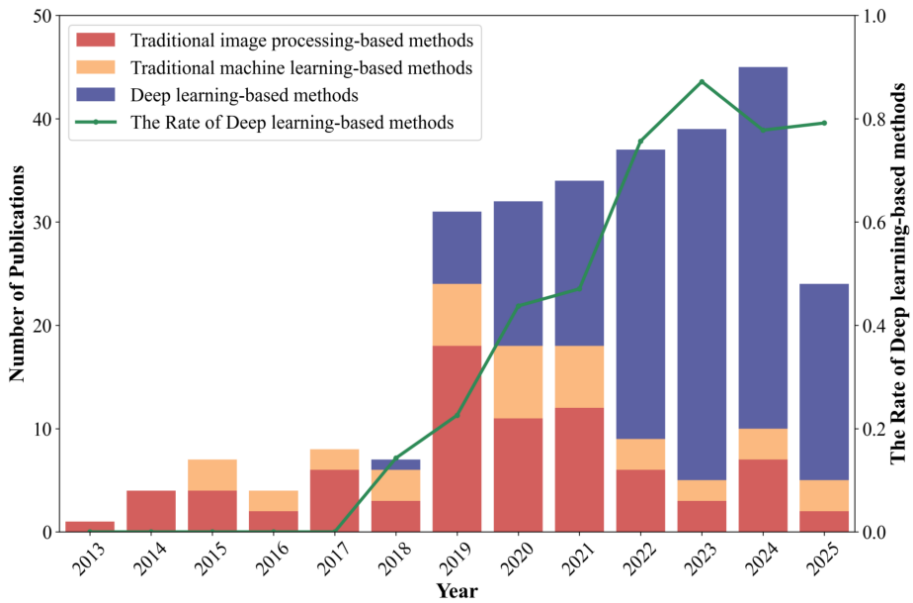
Foundation models such as Transformer architectures, the Segment Anything Model (SAM), and Vision-Language Vision-Language Models (VLMs), leveraging massive datasets and self-supervised learning, offer strong generalizaton ability, high adaptability, and improved performance accross diverse agricultural scenarios when applied to high-resolution remote sensing imagery.

The authors conceptualize APBD as composing three hierarchical levels:

1. Cropland Identification
2. Boundary Delineation
3. Parcel Segmentation

The paper seeks to synthesize the emerging trends in APBD research, presenting the historical evolution, current status, and future directions. The authors identify remote sensing data as a critical driver of APBD research in the foreseeable future.

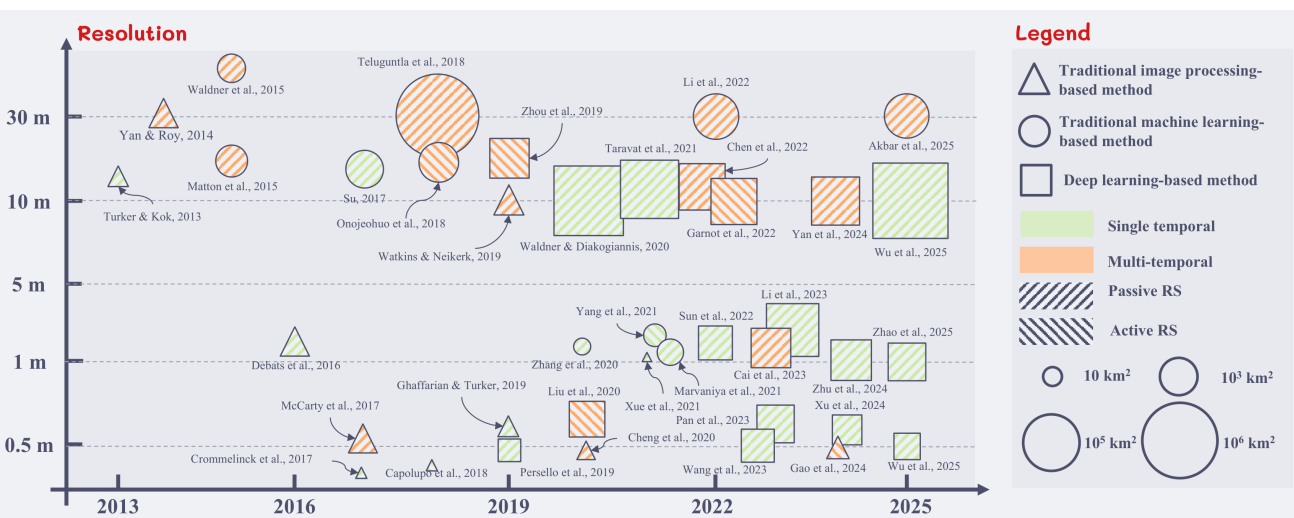
1.2 Meta-analysis of Related Literature



Trends in APBD methodologies (Figure 2)

I Overall Trend

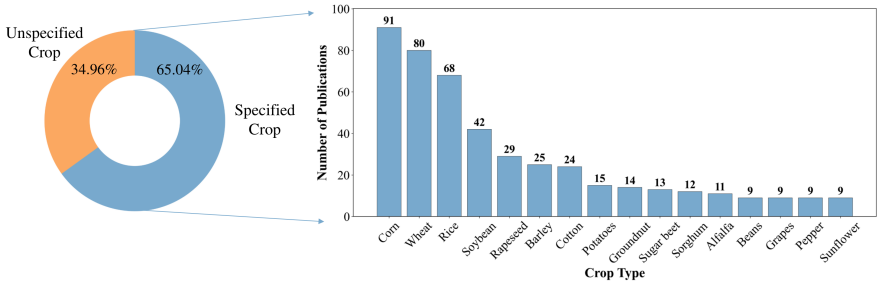
- *"Since 2019, the number of APBD studies using Very High-Resolution (VHR) imagery has increased exponentially."*
- Although VHR imagery enables high accuracy APBD, storage and processing costs prove to be a disadvantage. Large scale studies tend to use >10 m spatial resolution to balance land variety and coverage, accuracy, and storage.



Overall trends of APBD development (Figure 3)

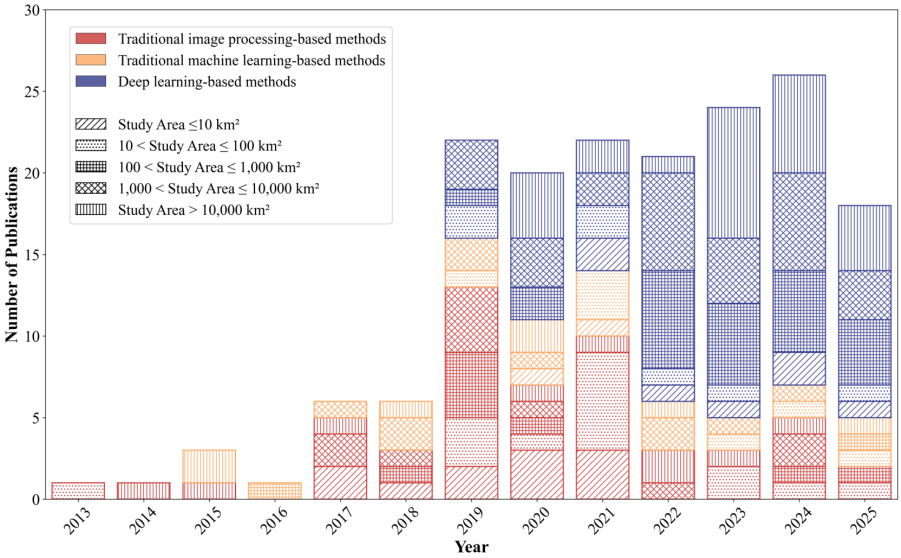
II Quantitative Analysis

- *Types of crops:* Corn crop most widely studied for APBD with statistics presented in figure 4 of the paper, attached below.



Statistics of crop types in APBD-related publications (Figure 4)

- *Study sites:* China (130 studies) leads in volume of publications followed by USA (19), Netherlands (14), and India (11).
- *Study area:* Rising trend in extremely large region studies (>10,000 km) catalysed by deep learning techniques.



Study area statistics for APBD publications (Figure 6)

- *Sensor type:* A strong reliance on satellite data (84.0%), followed by Unmanned Aerial Vehicles (10.8%) and aerial imagery (5.2%). Sentinel (82) and Landsat (31) lead the list in medium resolution satellite imagery, while GF series (62) is the overwhelming majority in high-resolution ($\leqslant$ 1m) studies.
- *Spatial resolution of data:* An increasing reliance on high-resolution imagery, attributed to the advent of modern deep learning algorithms that can extract detailed texture and spatial information.

III Methodology Review

Note: for most of the algorithms and methods mentioned below, the paper provides several good examples – these are not covered in the summary

A Traditional image processing-based methods

Traditional methods can be classified into:

1. *Pixel-based methods*: Each spatially continuous unit needs to be assigned a unique label.

- **Image binaryzation** classifies pixels belonging to a parcel using thresholds. Thresholds are determined globally or locally - based on neighboring pixels (the latter being more robust to high-value noise and low-value boundaries)
- **Clustering segmentation** groups similar pixels using clustering algorithms such as k-means, fuzzy c-means, etc.

2. *Edge-based methods*: Accurately detecting edges and boundaries between different parcels is a key challenge in APBD. Edge-based techniques are:
   - **Traditional edge detection ("gradient") operators, such as Canny and Sobel operators,** locate edge of the local area by using a differential operator to identify locations where the gray value abruptly changes in the image
   - **Wavelet transform** is useful for multi-resolution analysis and approximates well for 1-dimensional signals
   - **Globalized Probability of Boundary (gPb)** is an advanced edge detection algorithm that combines image segmentation, line detection, and contour generation integrating texture, color and brightness information at local and global scales
   - **Multiscale combinatorial grouping**
   - **Mean shift** is a nonparametric density estimation algorithm that eliminates the need to assume a sample distribution model and determine the number of categories, which can effectively reduce intra-field variation and preserve edge information
3. *Region-based methods*: start from the inside of an object and then expand outward until object boundaries are reached. (assumes that neighboring pixels within the same region/parcel have similar values)
   - **OBIA or GEOBIA** partitions remote-sensing imagery into meaningful image-objects which analyzes high spatial resolution imagery using spectral, spatial, topological, etc properties
   - **Watershed segmenation** is a mathematical, morphological segmentation algorithm that is based on topolgy theory and simulation of the process of water immersion, but is, however, prone to over-segmentation.
   - **Region growing** – a simple and popular algorithm
   - **Local binary fitting**

B Traditional machine learning-based APBD methods

The authors divide ML-based APBD into 4 steps:

1. image preprocessing
2. feature extraction
3. classifier training
4. model prediction

Acknowledgment

This report is based entirely on the work of Zheng et al. All concepts, findings, and frameworks presented herein are derived from their original research. No additional sources have been consulted for the core content of this summary.

References

Zheng, Juepeng et al. (2025). *A Comprehensive Review of Agricultural Parcel and Boundary Delineation from Remote Sensing Images: Recent Progress and Future Perspectives*. arXiv: 2508.14558 [cs.CV]. URL: https://arxiv.org/abs/2508.14558.